

Problem Set 6

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1 More combinatorics

In each problem, DO NOT just state the answer, explain how you came to the answer! For these problems, recall that we define the non-negative integers $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ and we define the positive integers $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$.

Problem 1: Prove for any positive integer k and all non-negative integer n ,

$$(x_1 + x_2 + \dots + x_k)^n = \sum_{b_1 + b_2 + \dots + b_k = n} \binom{n}{b_1, b_2, \dots, b_k} \prod_{j=1}^k x_j^{b_j}$$

To be clear on the notation, the summation is over all non-negative integer $b_1, b_2, \dots, b_k$ that sum to n and recall that $\binom{n}{b_1, b_2, \dots, b_k} = \frac{n!}{b_1! b_2! \dots b_k!}$.

Problem 2: Recall that there are $52! \approx 8.1 \times 10^{67}$ ways to order a standard deck of playing cards. How many distinct ways are there to order a game played by shuffling three such decks together? Here, treat the 52 cards of a single deck as distinct, but copies of the same card from different decks as identical.

Problem 3: How many non-negative integer solutions are there to the equation:

$$x + y + z = 10$$

Problem 4: How many positive integer solutions are there to the same equation from Problem 3? Make sure you explain the difference between the solutions of Problem 3 and 4.